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## Unit 4 : Session 3 : SLO 2 | Lambda Expressions and Anonymous Functions

### I. Programming Task : Custom Higher-Order Function

- `def transform(items, func):`
- `"""Apply func to every element of items and return the new list."""`
- `return [func(x) for x in items]`
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- `# usage`
- `nums = [1, 2, 3, 4]`
- `print(transform(nums, lambda x: x * x)) # [1, 4, 9, 16]`
- `print(transform(nums, lambda x: x + 10)) # [11, 12, 13, 14]`

### II. Explanation

- `transform()` is higher-order because it accepts another function (`func`) as a parameter.
- The lambda expressions are anonymous functions defined inline : no `def` or name needed.
- This pattern generalizes `map()`: any transformation can be plugged in without changing `transform()` itself.